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# TOWARDS AN IMPROVED HIGH-RISK PCI OUTCOMES



## CONSENSUS SUMMARY

### Article title

Clinical expert consensus document on the use of percutaneous left ventricular assist devices during complex high-risk PCI in India using a standardised algorithm

### Developed by

India Working Group (IWG) on high-risk PCI under the guidance and endorsement of the Indian Foundation of Advanced Cardiovascular Interventions, and two North American thought leaders; Dr. Rajiv Tayal (New Jersey) and Dr. Sanjog Kalra (Toronto)

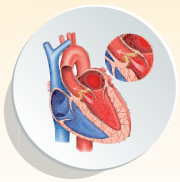
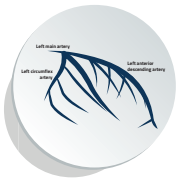
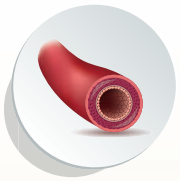
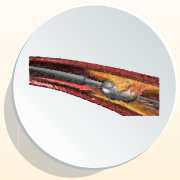
### Purpose

To guide the use of the Impella® pLVAD in high-risk PCI patients

**For healthcare professionals only.**

Complex and high-risk PCI patient population:

# How to define it?[1]

| FEATURES | IWG RECOMMENDATIONS |
|---|---|
| **CLINICAL ASPECTS** | |
|  Reduced left ventricular systolic function | Recommends the use of **Impella® pLVAD** in extensive revascularisation procedures performed on patients with severe LVSD. |
|  **Patient comorbidities:**<br>▪ Advanced age (> 75 years)<br>▪ Diabetes mellitus<br>▪ Advanced chronic kidney disease (GFR < 30 ml/min/1.73 $m^2$)<br>▪ Surgical turndown<br>▪ Valvular heart disease | ▪ Patients with diabetes or advanced kidney disease, with a higher risk of more complex, diffusely calcified multivessel disease, may benefit from protective devices.<br>▪ The use of **Impella® pLVAD** may be warranted in patients with valvular diseases with concomitant LVSD.* |
| **TECHNICAL ASPECTS** | |
|  UPLM stenosis or UPLM with distal disease (bifurcation or trifurcation) with a high SYNTAX score and concomitant LV dysfunction | The use of **Impella® pLVAD** may be beneficial. |
|  Multivessel disease with a large area of myocardium at risk and requirement for complete revascularisation (i.e. residual SYNTAX scores < 8) | If the multivessel disease involves the LAD–S1, or the LCx to OM, or the RCA to AM supplying a large area of myocardium then these procedures can be considered as high-risk PCI, and the use of an **Impella® pLVAD** support may be justified. |
|  Only remaining conduit artery (ORCA) | In cases requiring ORCA intervention or the use of a bypass graft ORCA as a conduit for recanalisation of the native coronary artery, the use of an **Impella® pLVAD** should be considered. |
|  Procedural characteristics | The use of **Impella® pLVAD** support in patients with moderate-to-severe LV dysfunction, which requires extensive or multi-territory atherectomy or the use of retrograde techniques for CTO recanalisation, should be considered. |

*__Disclaimer__: Contraindications to Impella® usage include mechanical aortic valves, severe aortic valvular stenosis or valvular regurgitation.[2]

# Decision-making tree for the use of haemodynamic support devices[1]

**RISK ASSESSMENT FOR HIGH-RISK PCI**

**ASSESSMENT BY MULTIDISCIPLINARY HEART TEAM AND SCORING SYSTEM***

**CLINICAL ASPECTS**

**Haemodynamic Status**

| | |
|---|---|
| LVEF <35% | 1 |
| LVEF <25% | 2 |
| SBP <100 mmHg | 1 |
| LVEDP >25 mmHg | 1 |
| Cardiac index <2 | 1 |

**Patient Characteristics**

| | |
|---|---|
| Renal Dysfunction | 1 |
| Advanced Age >75 yrs | 1 |
| ACS | 1 |
| Valvular Heart Disease (severe AS/MR) | 1 |
| Turned down for surgery | 1 |

**TECHNICAL ASPECTS**

**Coronary Anatomy**

| | |
|---|---|
| Unprotected left main disease<br>▪ Syntax score ≥32<br>▪ Syntax score <32 with bifurcation or trifurcation lesion | 1 |
| MVD with large area of myocardium at risk | 1 |
| Only remaining conduit vessel | 2 |

**Procedural Characteristics**

| | |
|---|---|
| Left main or multivessel atherectomy | 1 |
| Complex CTO with retrograde approach through last remaining conduit vessel | 1 |

| SCORE <3 | SCORE 3-5 | SCORE >5 |
|---|---|---|
| Conventional PCI | Consider support | Strongly consider support |

Proposed algorithmic risk assessment for protected PCI.
*Adapted from R. Tayal, S. Kalra, A. Seth et al; AsiaIntervention. 2022 Oct.*

The use of **Impella® pLVAD** support should be considered for a total score of > 5[1]

# Weaning of pLVAD[1]

**Post-procedure assessment**

**Weaning**

| Stable haemodynamics | Low-dose inotrope requirement | High-dose inotrope requirement |
|---|---|---|
| Wean/explant in the lab | Support in ICU for 6-24 hours | |
| | Wean/explant in ICU | Unable to wean in ICU |
| | | Upgrade to **Impella®** 5.0/5.5 or ECMO + **Impella®** |

Algorithm for weaning of pLVAD after completion of procedure.
*Adapted from R. Tayal, S. Kalra, A. Seth et al; AsiaIntervention. 2022 Oct.*

# Key highlights:

The **IWG Impella® patient selection tool and the scoring system** may assist interventional cardiologists in providing **safer and more appropriate revascularisation** to their highest risk patients[1]

Developed to ensure that the **needs of patients** are **optimally managed**, the algorithm has factored in the **economic constraint** faced by the patients[1]

The algorithm is not only **relevant** to be used in India, but also for other countries in **Asia**[1]

To provide patients with more complete revascularisation and optimised clinical outcomes, **choose Impella® pLVAD**

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**PCI**: percutaneous coronary intervention; **IWG**: India Working Group; **pLVAD**: percutaneous left ventricular assist device; **LVSD**: left ventricular systolic dysfunction; **GFR**: glomerular filtration rate; **UPLM**: unprotected left main coronary artery; **LV**: left ventricular; **LAD**: left anterior descending; **S1**: 1st septal; **LCx**: left circumflex; **OM**: obtuse marginal; **RCA**: right coronary artery; **AM**: acute marginal; **ORCA**: only remaining conduit artery; **CTO**: chronic total occlusion; **LVEF**: left ventricular ejection fraction; **SBP**: systolic blood pressure; **LVEDP**: left ventricular end diastolic pressure; **ACS**: acute coronary syndrome; **AS**: aortic stenosis; **MR**: mitral regurgitation; **MVD**: multivessel disease; **ICU**: intensive care unit; **ECMO**: extracorporeal membrane oxygenation.

**References:**


1. Tayal R, Kalra S, Seth A, *et al.* Clinical expert consensus document on the use of percutaneous left ventricular assist devices during complex high-risk PCI in India using a standardised algorithm. *AsiaIntervention.* 2022;8(2):75-85.
2. Ergle K, Parto P, Krim SR. Percutaneous ventricular assist devices: a novel approach in the management of patients with acute cardiogenic shock. *Ochsner J.* 2016;16(3):243-9.


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HRPCI-APACJan2023